

BCL Barcelinhos Autonomous Rescue Simulation Robot

Team Barcelinhos ▪ RoboCup Junior Rescue Simulation 2026 ▪ Webots / Erebus Platform ▪ Pure Python 3, no external libraries

Team

Catarina — Navigation & Mapping
Mariana — Perception & Sensors
Mentors
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System at a Glance

EKF sensor fusion ▪ Log-odds mapping
Priority reactive nav ▪ Triple anti-stuck
Ring-sum hazmat ▪ Hole avoidance

Objectives

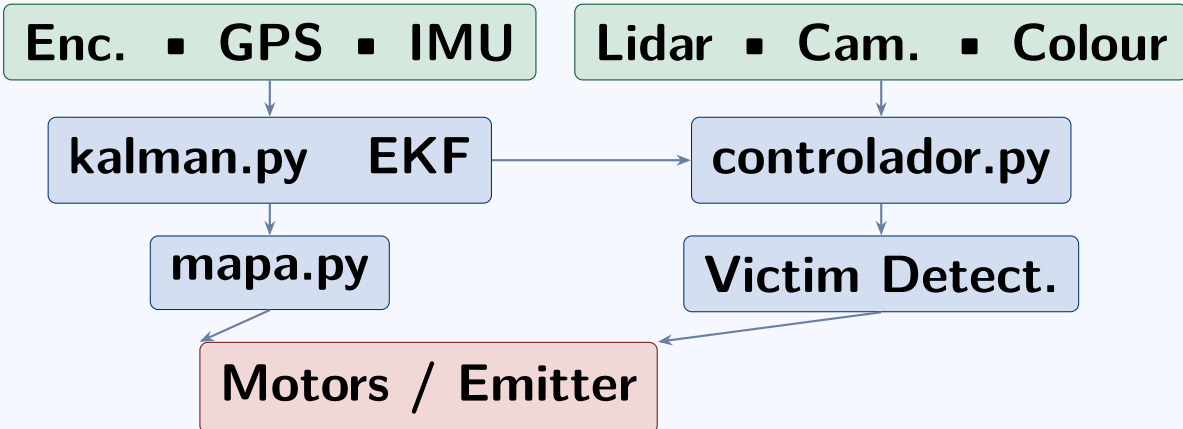
- Autonomously **navigate** an unknown maze.
- Build a live **2-D occupancy map** (log-odds, 5 cm/cell).
- Detect & report** victims (Φ, Ψ, Ω) and hazmats (F, P, C, O).
- Avoid holes** via floor colour sensor.
- Recover from blockages via **triple anti-stuck**.

Hardware — Erebus_Bot

Sensor / Actuator	Role
Lidar 360° — 512 pts, 4 layers	Nav. + Mapping
GPS	KF position
IMU (yaw)	KF heading
Wheel encoders	KF odometry
Camera 64×64 px	Victim / hazmat ID
Floor colour sensor	Hole detection
Emitter	Victim reporting

$r=20\text{ mm}$ $b=52\text{ mm}$ $v_{\text{max}}=6.28\text{ rad/s}$ $\Delta t=32\text{ ms}$

Software Architecture



Triple Anti-Stuck System

GPS stuck (50 steps) $\Delta\text{pos} < 2\text{ mm/step}$ **Encoder stuck (55 steps)** $\text{arc} < 0.05\text{ rad}$ **Spinning (60 steps)** wheels move, GPS static

do_escape(): reverse 12 cm → rotate → advance 4.9 cm

Rotation: $90^\circ \pm \{0^\circ, 45^\circ, 90^\circ, 135^\circ\}$ (cycles to avoid loops). 100-step cooldown after each escape.

Extended Kalman Filter — Pose Estimation

State: $\mathbf{x} = [x, z, \theta]^\top$ position in metres, heading in radians

Predict — differential-drive kinematics:

$$d_c = \frac{r(\delta_L + \delta_R)}{2}, \quad \Delta\theta = \frac{r(\delta_R - \delta_L)}{b}$$
$$\hat{\mathbf{x}} = \left[x + d_c \cos\left(\theta + \frac{\Delta\theta}{2}\right), \quad z + d_c \sin\left(\theta + \frac{\Delta\theta}{2}\right), \quad \theta + \Delta\theta \right]^\top$$

Update — measurement models (every 32 ms):

GPS R = diag(0.04, 0.04) — tolerates Erebus GPS noise ($\sigma \approx 20\text{ cm}$)

IMU yaw R = $5 \times 10^{-3}\text{ rad}^2$ — high heading confidence

Q = diag(4×10^{-4} , 4×10^{-4} , 10^{-4})

Fuses noisy GPS + precise IMU + encoder odometry.
 $\sigma_{\text{pos}} \approx 8\text{--}15\text{ cm}$ — feeds mapping, anti-stuck & victim deduplication.

Navigation — Priority Behaviours (P1–P4)

Each timestep, conditions are tested top-to-bottom; the first match fires and lower priorities are skipped.

P1 — Wall stop front $< 0.15\text{ m}$ or turn locked → **Spin on axis**

P2 — Side push Side urgent & front $< 0.30\text{ m}$ → **Differential lateral push**

P3 — Steer front $< 0.25\text{ m}$ or hold active → **Steer while advancing**

P4 — Clear No obstacle ahead → **Full speed + heading lock**

Heading lock (P4): $c = \text{clamp}(4.0e, -2.0, +2.0)\text{ rad/s}$, $e = \theta_{\text{target}} - \theta_{\text{KF}}$ (wrapped to $(-\pi, \pi]$).

Corridor mode: When the front sensor reads clear, lateral push (P2) is suppressed and asymmetric fr_l / fr_r readings steer the robot through narrow passage entries without overcorrecting.

Sensor thresholds:

D_FRONT_STOP	0.15 m
D_FRONT_TURN	0.25 m
D_SIDE_PUSH	0.12 m
D_SIDE_ACTIVE	0.30 m

Occupancy Grid Mapping

Grid: 120×120 cells, 5 cm resolution, $6\text{ m} \times 6\text{ m}$.

Log-odds update per lidar ray (512 rays, every 10 steps):

free cells: $\ell += -0.4$
endpoint: $\ell += +0.9$, $\ell \in [-4, +4]$
 $\ell > 0.5 \Rightarrow$ **obstacle**
 $\ell < -0.2 \Rightarrow$ **free / explored**
 $\ell \approx 0 \Rightarrow$ unknown

Noise dynamics: ratio $0.9/0.4 = 2.25$: spurious hits clear in ≤ 3 passes ($< 1\text{ s}$); walls saturate to $\ell=4.0$ and persist stably.

Submission: down-sampled to 60×60 at 10 cm/cell; binary payload via emitter.

Victim & Hazmat Detection

Camera frame ($64 \times 64\text{ px}$), every 25 steps. Two classifiers run in parallel:

Hazmat (colour $> 10\%$ & white $> 5\%$)

5 concentric rings, 8 angles each.
Ring-sum: $\{B=-2, R=-1, Y=0, G=+1, Bl=+2\}$
 $\Sigma=0 \rightarrow F, 1 \rightarrow P, 2 \rightarrow C, 3 \rightarrow O$

Letter victim (white $> 45\%$ & black $> 8\%$)

Region-density: $s(\Phi)=d_L+d_R-0.5d_C$, $s(\Psi)=d_C+d_B$, $s(\Omega)=d_T-0.5d_B$
Highest score → reported as H

Report: `struct.pack('i i c', x_cm, z_cm, char)` — 9 bytes (Erebus format).

Dedup: `(round(x×12), round(z×12))` hash.

Hole Avoidance & Key Innovations

Detection: floor brightness $< 25/255$.

Escape: stop → reverse 12 cm → rotate 180° (IMU loop) → navigate away (80-step cooldown).

Key Innovations

- ★ Full EKF in pure Python — no external libs.
- ★ Log-odds ratio 2.25: noise clears $< 1\text{ s}$; walls saturate at $\ell=4$.
- ★ Startup lidar diagnostic selects the horizontal scan layer.
- ★ Adaptive escape angles — never repeats the same rotation.
- ★ $O(1)$ ring-sum hazmat classifier, zero training data.

$v_{\text{fwd}}=5.5\text{ rad/s}$ $\Delta t=32\text{ ms/step}$
Map: $6 \times 6\text{ m}$ at 5 cm/cell
Lidar: 512 pts, 4 layers, 1 m range
Camera: $64 \times 64\text{ px}$ / 25 steps
EKF: GPS + IMU + encoders